IV Infusions Practice 1

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

Continuous intravenous infusion of a potent analgesic is recommended for a patient with inoperable colon cancer and a body weight of 110 pounds. The dose is prepared by dissolving 200 mg of the drug in 500 mL of 5% dextrose. The drug is to be infused over 24 hours. The drug has a half-life of 4 hours and an apparent volume of distribution of 3 L/kg. Calculate the following:

a Rate of drug infusion in mg/hr

b Plasma concentration 6 hours after the start of the infusion

c Total amount of drug in the body 12 hours after the start of the infusion

d Steady-state plasma concentration

e Rate of drug infusion in mL/min

f An IV bolus loading dose

g Plasma concentration 12 hours after the cessation of the infusion

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 8.33 mg/hr

WHAT THE STEM GIVES

weight	110 pounds	*not needed for the rate*
D_0	200 mg	the numerator
volume	500 mL	*not needed for mg/hr; used in part e*
t inf	24 hours	the denominator
t½	4 hours	*not needed for the rate*
V_D	3 L/kg	*not needed for the rate*
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{D_0}{t\ inf}$$

A constant infusion delivers the dose evenly over the infusion time, so the rate is dose over time.

ALGEBRA

$$R = \frac{200\ mg}{24\ hr} = 8.33\ mg/hr$$

mg over hr gives mg/hr, the amount entering the body each hour. The 500 mL of dextrose is the volume the drug is dissolved in and does not change the rate in mg.

> **DOES THE NUMBER MAKE SENSE?**
> 200 mg over a full day is a little over 8 mg each hour; 8.33 mg/hr fits.

b

Her key

Answer: 0.207 mg/L

WHAT THE STEM GIVES

weight	110 pounds	$\frac{110}{2.2} = 50$ kg
D_0	200 mg	gave R in part a
volume	500 mL	*not needed for a plasma concentration*
t inf	24 hours	gave R; the infusion still runs at 6 hours
t½	4 hours	$k = \frac{0.693}{4} = 0.1733\ hr^{-1}$
V_D	3 L/kg	times 50 kg gives 150 L
R	8.33 mg/hr	the numerator of the plateau

t	6 hours	the exponent kt
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

UNIT

$$W = \frac{110 \text{ lb}}{2.2 \text{ lb/kg}} = 50 \text{ kg}; \; V_D = (3 \text{ L/kg})(50 \text{ kg}) = 150 \text{ L}$$

Pounds over lb/kg leaves kilograms, and L/kg times kg leaves litres.

SETUP

$$C_p = \frac{R}{kV_D}(1 - e^{-kt}), \text{ with } k = \frac{0.693}{4 \text{ hr}} = 0.1733 \text{ hr}^{-1}$$

During a constant infusion the level rises toward the plateau $\frac{R}{kV_D}$; $(1 - e^{-kt})$ is the fraction of the plateau reached by time t.

ALGEBRA

$$C6 = \frac{8.33 \text{ mg/hr}}{(150 \text{ L})(0.1733 \text{ hr}^{-1})}(1 - e^{-(0.1733)(6)}) = (0.3204 \text{ mg/L})(0.6465) = 0.207 \text{ mg/L}$$

L × hr^{-1} = L/hr, and mg/hr over L/hr gives mg/L, the plateau. kt has no units; 64.65% of the plateau is reached at 6 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is one and a half half-lives, so the level is between half and three quarters of the 0.3204 mg/L plateau: 0.207 mg/L fits.

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c Her key

Answer: 42 mg

WHAT THE STEM GIVES

weight	110 pounds	$\frac{110}{2.2} = 50$ kg
D_0	200 mg	gave R in part a
volume	500 mL	*not needed*
t inf	24 hours	the infusion still runs at 12 hours
t½	4 hours	k = 0.1733 hr^{-1}; 12 hours is three half-lives
V_D	3 L/kg	times 50 kg gives 150 L
R	8.33 mg/hr	the numerator of the plateau
t	12 hours	the exponent kt
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \frac{R}{kV_D}(1 - e^{-kt}), \text{ then } D_B = C_p \times V_D$$

The concentration at 12 hours comes from the during-infusion line; the amount in the body is that concentration times the volume of distribution.

ALGEBRA

$$C12 = \frac{8.33 \text{ mg/hr}}{(150 \text{ L})(0.1733 \text{ hr}^{-1})}(1 - e^{-(0.1733)(12)}) = (0.3204 \text{ mg/L})(0.875) = 0.28 \text{ mg/L}$$

mg/hr over L/hr gives mg/L. Twelve hours is three half-lives, so 87.5% of the plateau is reached.

ALGEBRA

$$D12 = (0.28 \text{ mg/L})(150 \text{ L}) = 42 \text{ mg}$$

mg/L times L cancels the litres and leaves mg, the amount of drug in the body at 12 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is three half-lives, so the body holds 87.5% of the amount it would hold at the plateau, 0.3204 mg/L in 150 L; 42 mg fits.

Carried unrounded, the amount is 42.1 mg; her key rounds the concentration to 0.28 mg/L first and gets 42 mg. Both are accepted.

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d

Her key

Answer: 0.32 mg/L

WHAT THE STEM GIVES

weight	110 pounds	$\frac{110}{2.2} = 50$ kg
D_0	200 mg	gave R in part a
volume	500 mL	*not needed*
t inf	24 hours	gave R
t½	4 hours	$k = \frac{0.693}{4} = 0.1733 \text{ hr}^{-1}$
V_D	3 L/kg	times 50 kg gives 150 L
R	8.33 mg/hr	the numerator of C_{ss}
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{ss} = \frac{R}{kV_D} = \frac{R}{Cl_T}$$

At steady state the rate in equals the rate out, $R = Cl_T \times C_{ss}$, so the plateau is the infusion rate over clearance, $Cl_T = kV_D$.

ALGEBRA

$$C_{ss} = \frac{8.33 \text{ mg/hr}}{(0.1733 \text{ hr}^{-1})(150 \text{ L})} = 0.32 \text{ mg/L}$$

$hr^{-1} \times L = L/hr$, here (0.1733)(150) = 26 L/hr, and mg/hr over L/hr leaves mg/L: the hours cancel.

> **DOES THE NUMBER MAKE SENSE?**
> The plateau is the rate over clearance, 8.33 mg/hr over (0.1733)(150) L/hr, about a third of a mg in each litre; 0.32 mg/L fits.

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e

Her key

Answer: 0.35 mL/min

WHAT THE STEM GIVES

volume	500 mL	the numerator
t inf	24 hours	times 60 minutes, the denominator
D_0	200 mg	*not needed: a volume rate is asked*
weight	110 pounds	*not needed*
t½	4 hours	*not needed*
V_D	3 L/kg	*not needed*
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Rate} = \frac{\text{volume}}{\text{time}}$$

The 500 mL of solution runs in evenly over 24 hours, so the volume rate is the volume over the infusion time.

UNIT

$$\text{Rate} = \left(\frac{500 \text{ mL}}{24 \text{ hr}}\right)\left(\frac{1 \text{ hr}}{60 \text{ min}}\right) = 0.35 \text{ mL/min}$$

mL/hr times hr/min: the hours cancel and mL/min is left. $\frac{500}{1440} = 0.347$ mL/min, reported as 0.35.

> **DOES THE NUMBER MAKE SENSE?**
> 500 mL over 24 hours is about 21 mL each hour, a third of a mL each minute; 0.35 mL/min fits.

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f

Her key

Answer: 48 mg

WHAT THE STEM GIVES

weight	110 pounds	*not needed: R over k needs no volume*
D_0	200 mg	gave R in part a

t½	4 hours	$k = \frac{0.693}{4} = 0.1733\ hr^{-1}$, the denominator
V_D	3 L/kg	*not needed by R over k; $C_{ss} \times V_D$ gives the same*
R	8.33 mg/hr	the numerator
volume	500 mL	*not needed for the loading dose*
t inf	24 hours	gave R in part a
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = \frac{R}{k}$$

A loading dose puts the plateau amount in the body at once; that amount is the infusion rate over the elimination rate constant, which equals $C_{ss} \times V_D$.

ALGEBRA

$$D_L = \frac{8.33\ mg/hr}{0.1733\ hr^{-1}} = 48\ mg$$

mg/hr over hr^{-1}: the hours cancel and mg is left. $\frac{8.33}{0.1733} = 48.07$, reported as 48 mg.

> **DOES THE NUMBER MAKE SENSE?**
> The loading dose is the amount the body holds at the plateau: 8.33 mg/hr over 0.1733 hr^{-1}, about 48 mg, a quarter of the 200 mg infused over 24 hours.

With the unrounded rate, 8.333 mg/hr, the loading dose is 48.1 mg, which is also $C_{ss} \times V_D$ = (0.3207 mg/L)(150 L); her key gives 48 mg. Both are accepted.

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g

Her key

Answer: 0.04 mg/L

WHAT THE STEM GIVES

t inf	24 hours	six half-lives: the level is at its plateau when the infusion stops
t½	4 hours	k = 0.1733 hr^{-1}; 12 hours is three half-lives
C_{ss}	0.32 mg/L	the starting concentration of the decline
t	12 hours	after the cessation: the exponent kt
weight	110 pounds	*not needed: C_{ss} is given*
V_D	3 L/kg	*not needed: C_{ss} is given*
D_0	200 mg	gave R and C_{ss}; not needed once C_{ss} is given
volume	500 mL	*not needed*
dextrose	5%	*not needed: the diluent the dose is dissolved in*

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = C_{ss}\, e^{-kt}$, with t counted from the end of the infusion

After 24 hours, six half-lives, the infusion has brought the level to its plateau; once it stops nothing enters, and the level falls first order from there.

ALGEBRA

$C12 = (0.32\ mg/L)\, e^{-(0.1733)(12)} = (0.32\ mg/L)(0.125) = 0.04\ mg/L$

kt = 2.08 has no units, so the answer keeps the mg/L of C_{ss}; $e^{-2.08} = 0.125$ is the fraction left.

ALGEBRA

12 hr is 3 half-lives: 12.5% remains, (0.125)(0.32 mg/L) = 0.04 mg/L

Her key also counts half-lives: at 12 hours, three half-lives, 12.5% of the starting concentration remains.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is three half-lives of 4 hours, so one eighth of the 0.32 mg/L remains: 0.04 mg/L.

Her key starts the decline from C_{ss}, 0.32 mg/L. After 24 hours the infusion has reached 98.4% of C_{ss}, 0.3157 mg/L, which gives 0.0395 mg/L at 12 hours; both round to 0.04 mg/L and both are accepted.